

Implementation of Peer-to-Peer Energy Trading via Solana Smart Contracts in a Permissioned Environment

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Thailand's National Energy Trading Platform (NETP), envisaged since 2017 as a blockchain settlement layer for Peer-to-Peer (P2P) trading among rooftop-solar prosumers, remains in development, and the regulated buy-back still gives prosumers no direct mechanism to exchange surplus generation within the local distribution grid. This paper presents an on-chain settlement layer for P2P energy trading: six modular Anchor programs on a permissioned Solana runtime, where routing every high-frequency write to a per-entity program-derived account (PDA) lets transactions with disjoint write sets execute in parallel under Sealevel, ordered by Proof-of-History (PoH). The market-clearing framework couples a continuous double auction (CDA) for real-time trades with a uniform-price auction clearing aggregated bids at a single price over fifteen-minute epochs. We simulate 80 meters over 30 days at fifteen-minute intervals on a low-voltage feeder, 15% of them prosumers, and evaluate the full token life-cycle on a single-node validator; the community-month replays 1,858× faster than real time with zero delivery loss. Three findings follow. Primarily, settlement is bound by packet framing rather than by compute: a match leaves about half the 200 k default compute budget unused, yet the 1,232-byte packet admits only one match, and Address Lookup Tables cannot relieve it. Denser settlement needs signatures repackaged, not more compute — a constraint transferable to any signature-verifying settlement layer. Second, settlement is exactly auditable: fourteen chain-derived conservation assertions all close. Third, participation is decided by the price rule rather than the platform, the wheeling charge ranking the uniform-price auction above the regulated buy-back and the CDA.

1. Motivation and Design Rationale

Thailand's state utilities have advanced a National Energy Trading Platform (NETP) since 2017 to facilitate P2P trading among rooftop-solar prosumers, using a Tendermint-based consortium blockchain [1]. This article presents a Solana-based architecture as an alternative execution layer, motivated by NETP's own finding that Tendermint outperformed Ethereum and Hyperledger Fabric in throughput testing, which suggests that further gains may be available from Solana's parallel-execution model. P2P market design and blockchain-based energy trading are already well surveyed [2], [3] and demonstrated in field pilots such as the Brooklyn Microgrid [4]. What such work reports is throughput and compute cost, presuming computation to be the axis along which a settlement layer scales. This paper reports the constraint that binds once it is not, with the conservation evidence throughput figures alone cannot give. The settlement logic is built as six programs on Anchor, Solana's official framework, written in Rust, compiled to SBF bytecode, and executed on the Solana Virtual Machine (SVM). By routing high-frequency writes to per-entity program-derived accounts (PDAs), transactions with disjoint write sets never contend for the same account lock and therefore execute in parallel under Solana's Sealevel runtime. The design claim this paper tests is what remains once that partitioning removes computation as the limit: consensus and lock serialization bound throughput, and transaction size, not compute cost, bounds settlement density.

2. On-Chain Settlement Layer and Simulated Grid

The evaluated system has two parts, the settlement layer under test and the simulated grid that drives it:

- The smart-contract layer is built on the Solana Anchor framework [5] (anchor-lang and anchor-spl, version 1.0.0) and comprises six programs, **energy-token**, **governance**, **oracle**, **registry**, **trading**, and **treasury**, all written in Rust and compiled to SBF bytecode. One program invokes another through cross-program invocation (CPI), with authority delegated to program-derived addresses (PDAs) that sign without any private key existing. The deployment targets a permissioned environment realized as a private solana-test-validator network; all experiments reported here ran on a single node (Apple M2, Agave 3.1.10).
- The fleet uses the CINELDI 80-bus rural LV reference grid (SINTEF) [6] for network topology only, the 80-node single-phase feeder with its V/A/W parameters, resolved for power flow with pandapower [7]. As CINELDI carries only hourly load and no generation, all time-series are simulator-produced: per-meter consumption and twelve prosumer nodes' (15%) rooftop PV at fifteen-minute cadence, with solar modeled deterministically via pvlib [8] under a fixed seed.

A complete run must satisfy three closure identities, each checkable from chain state alone:

$$\sum_{i,d} M(i,d) = E_{\text{settle}} = E_{\text{burn}}, \quad \sum_i R(i) = \frac{\text{REC supply}}{10^3}, \quad \sum_m T_m = P_{\text{sell}} + W \quad (1)$$

Here $M(i,d)$ is the GRID minted for prosumer i on day d ; E_{settle} and E_{burn} the energy moved by settlement and retired by burns; $R(i)$ the REC certified to i ; and T_m , P_{sell} , W the per-match buyer debit, seller proceeds, and wheeling charges. An audit harness re-derives the fourteen assertions behind Eq. (1) from live chain state by RPC reads alone: per-meter counters, PDA censuses, registry conservation sums, token supplies and balances.

3. Throughput, Audit, and Revenue Results

Throughput is client-observed confirmed goodput, confirmed transactions per wall-clock second from burst start to last confirmation. Every non-confirmed transaction is attributed to a failure class (submission reject, on-chain guard, or expiry), separating delivery loss from validation working as intended. Table 1 gives the workload every reported figure derives from.

Table 1: The simulated community-month dataset (seed-deterministic; no field data).

Quantity	Value
fleet	80 meters (seeded solar/load physics): 12 prosumer sellers, 68 consumers
market policy	sell cap $C = 10$ kWh/day per prosumer; 0.1 kWh dust floor
horizon / readings	30 days $\times$ 96 ticks ($\Delta t = 15$ min) = 230,400 readings, integer Wh
month energy	15,868.5 kWh generated; 144,879.8 kWh consumed; 10,386.7 kWh interval surplus
oracle-accepted surplus	8,253.502 kWh; 919 readings gate-rejected

3.1. Measured Outcomes and Limitations

Table 2 reports the replay. The first two findings are technical. Compute does not limit settlement: a match spends 107 k CU, about half the 200 k default budget, yet only one match fits per transaction. The 1,232-byte packet binds instead, since each match carries its 77-byte signed order twice, in the Ed25519 verification instruction and again as settlement instruction data, so a pair costs 186 bytes and a second pair overruns the packet. Address Lookup Tables cannot recover it, compressing account keys rather than instruction data. Second, settlement is exactly auditable, as Eq. (1) holds on chain under all three price rules.

Table 2: Community-month replay outcomes, each figure re-derived from on-chain state.

Metric	Value	What it bounds
replay wall-clock	1,394.7 s for 230,400 readings (1,858 $\times$ real time)	end-to-end lifecycle cost
telemetry ingest	≈ 213 readings $\cdot$ s ⁻¹	oracle write path under PDA partition
order bursts	≈ 53 TPS confirmed goodput	trading account-lock serialization
settlement	107 k CU/match ($\approx 54\%$ of the 200 k budget); 1 match/tx	1,232-byte packet, not compute
delivery loss	0 unconfirmed; 919 readings gate-rejected	validation, not lost throughput
audit	14/14 chain-derived assertions	exactness of settlement

Goodput on a single validator bounds execution and locking only, not consensus in a three-utility consortium. The telemetry outcome reproduced bit-identically across four runs, throughput varying 213–232 readings $\cdot$ s⁻¹, while the lifecycle figures come from one canonical run and the price-rule comparison from a separate sweep, since nullifier PDAs make identical replay impossible on a persistent ledger. The dataset is simulator-produced, with no field measurements.

3.2. Where the Prosumer’s Revenue Goes

The third finding is economic. All three rules move identical energy in Table 3, so the ranking is set by the deductions: wheeling takes 1,278 THB from each P2P rule, a third of the uniform auction’s gross and more of the CDA’s, and nothing from the buy-back, while fees (25 bp) and the loss allowance (5 bp) together take about 0.3%. The uniform auction survives wheeling and still beats the buy-back; the CDA does not, and the ranking reverses once supply is left uncapped. Wheeling policy is the regulator’s lever.

Table 3: Prosumer revenue decomposition for one cleared month (1,111.4 kWh; 80 meters / 8 prosumers, 30 days), in THB.

Price rule	Gross	Fee	Wheeling	Loss	Net	Net THB/kWh
uniform	3,834.31	9.59	1,278.10	1.92	2,544.70	2.290
buy-back	2,445.07	0.00	0.00	0.00	2,445.07	2.200
CDA	3,584.25	8.96	1,278.10	1.79	2,295.39	2.065

Conclusion

The measurements support the central design claim and sharpen it: with hot-path state in per-entity PDAs computation never binds, consensus and lock serialization bound throughput, and packet framing bounds how much settlement a transaction carries, while settlement stays exactly auditable. What remains open is economic: the wheeling charge, not the ledger, sets the threshold at which a Thai prosumer prefers P2P trading to the buy-back. A multi-validator consortium mapped to EGAT, MEA, and PEA is the natural next step, suited to UEC–ASEAN collaboration.

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